Chapter 2

Low Level Image Descriptors: An Overview

We begin with an overview of the low level color and texture image feature descriptors. The overview is intended to provide the background knowledge for the descriptors investigated in the following chapters. For the texture feature, a homogeneous texture descriptor is introduced in detail; The computation of the descriptor also provides the framework based on which the texture browsing descriptor presented in Chapter 3 is computed. For the color feature, the quantized color histogram descriptor is presented. Both the color and texture features have been widely used in similarity retrieval applications. Using the state of art descriptors offers a realistic environment to address the relevant issues in the following chapters.

2.1 Introduction

The low level features provide a fundamental means to characterize image/video objects. Some typical features used to describe image objects include color, texture and shape. For video objects, besides the features for image objects, descriptors that characterize motion and temporal information have been investigated. More than one descriptors can be used to characterize a feature. For instance, in the upcoming MPEG-7 standard [2], three descriptors are specified to characterize texture feature: the homogeneous texture descriptor, texture browsing descriptor, and the edge histogram descriptor. The color feature in MPEG-7 is character-

ized by the scalable color histogram, color structure histogram, dominant color, and color layout.

From an image/video analysis point of view, each of the descriptors mentioned above has been researched extensively. The research usually focuses on two related issues. One is about the effectiveness of the descriptor for describing its intended attribute. For example, the dominant color descriptor characterizes a set of perceivable dominant colors in a given image region. The second issue is about the effectiveness of the descriptor in similarity based retrieval. A descriptor's overall effectiveness is typically evaluated in similarity retrieval context. Similarity retrieval evaluation, being a subjective process, often results in blurring the two issues -- the first one being about the syntax (what it represents) of the descriptor, whereas the second one relates to the semantics.

A collection of descriptors for a given set of media data constitute a low level representations that can be used for many different applications. In order to be effective, these basic descriptors need to be organized. Simple operations such as insertion or deletion of new records in a large database of such media descriptors can pose significant problems. Part of the problem is the high dimensionality of many of the popular low level descriptors for color and texture. For example, a typical color histogram can be a vector in 64 or even 256 dimensions; The dimensionality of the MPEG-7 homogeneous texture descriptor is 62. Such high dimensions can significantly affect the cost of indexing and updating the database. These issues are addressed in Chapter 4, 5, and 6.

In this chapter, we mainly focus on descriptors related with two image features, texture and color. In Chapter 2.2, a state of the art homogeneous texture descriptor is presented. Chapter 2.3 introduces the traditional color histogram descriptor briefly. The two presented descriptors are widely used to measure the similarity of image/video objects. Some examples of similarity image retrieval based on texture and color features are presented to illustrate the usage and the effectiveness of these descriptors.

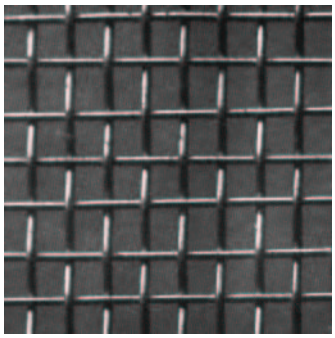
2.2 A Texture Descriptor for Similarity Retrieval

Image texture can be observed from a wide variety of image objects. For example, pictures of water, a pattern on a fabric, grass, etc., contain strong examples of image texture. A collection of image textures in [14] provides texture patterns ranging from man made textures (woven aluminum wire, brick walls, handwoven rugs, etc.) to natural objects (water, grass, clouds, etc.).

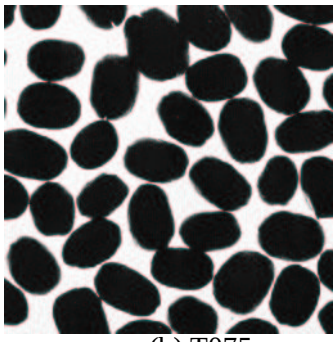
Though image texture is a widely used term in image analysis and computer vision, it dose not have a precise definition. In a very general sense, one can think of a texture as consisting of some basic primitives (micro-patterns), whose spatial distribution in the image creates the appearance of a texture. The spatial distribution of the micro-patterns could be regular or random. In Figure 2-1(a), the grid structure is a micro-pattern whose regular distribution constitutes a structured pattern of “Woven aluminum wire”. But such regular distribution of micro-pattern can not be observed from Figure 2-1(b). For many cases, such well-defined micro-patterns may not exist, see Figure 2-1(c) and (d).

Traditional research areas related with image texture analysis are texture classification, texture segmentation, and texture synthesis. In recent years, with the proliferation of image/video data, image texture has emerged as an important visual feature for image/video information retrieval applications. One common task involved in these applications is to measure the similarity of texture patterns. For example, in the *query by example* application, given a query image, one is interested in finding visually similar images from an image database by ranking the similarity of images with the query image. For this purpose, texture descriptors are derived to characterize the texture pattern and used to compute the similarity.

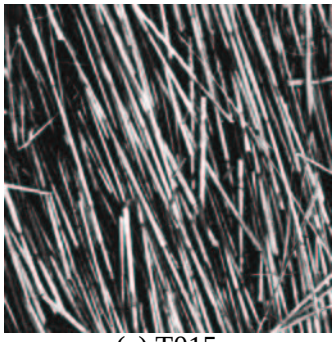
Several recently developed homogeneous texture descriptors for similarity retrieval include the descriptor based on multi-resolution simultaneous auto-regressive model (MRSAR) [54], a pyramid-structured wavelet transform (PWT) descriptor, tree-structured transform (TWT) based descriptor [17] and multi-resolution Gabor filtering based descriptor [52]. In this dissertation, the descriptor based on multi-resolution Gabor filtering is adopted as the descriptor. The main reason is that from recent evaluation [43], [51], [52], this descriptor compares favorably with other descriptors mentioned above in differentiating homogeneous



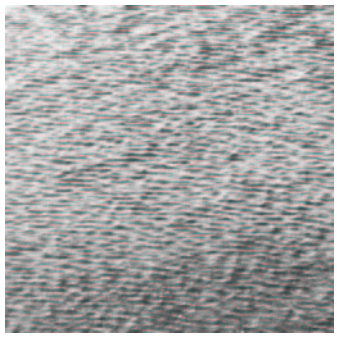
(a) T001



(b) T075



(c) T015



(d) T038

FIGURE 2-1. (a) Woven aluminum wire (T001); (b) Coffee beans (T075); (c) Straw (T015); (d) Water (T038).The numbers refer to orders of the textures in Brodadz album [14]

texture patterns. In this dissertation, we refer to this descriptor simply as the homogeneous texture descriptor (HTD).

In the following, we introduce the computation and utilization of HTD for similarity retrieval.

2.2.1 Gabor filter bank

Gabor filters can be considered as orientation and scale tunable edge and line detectors, and the statistics of these micro features can be used to characterize the underlying texture. A two dimensional Gabor function and its Fourier transform can be written as:

$$g(x, y) = \frac{1}{2\pi\sigma_x\sigma_y}\exp\left[-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2}+\frac{y^2}{\sigma_y^2}\right)+2\pi jWx\right] \tag{2-1}$$

$$G(u, v) = \exp\left\{-\frac{1}{2}\left[\frac{(u-W)^2}{\sigma_u^2}+\frac{v^2}{\sigma_v^2}\right]\right\} \tag{2-2}$$

where $\sigma_u = 1/(2\pi\sigma_x)$ and $\sigma_v = 1/(2\pi\sigma_y)$. Using $g(x, y)$ as the mother wavelet, a bank of filters can be derived by appropriately dilating and translating the $g(x, y)$ through the generation function:

$$\begin{aligned} g_{mn}(x, y) &= a^{-m}g(x', y') \qquad a > 1 \qquad m, n = intege \\ x' &= a^{-m}(x\cos\theta + y\sin\theta) \qquad y' = a^{-m}(-x\sin\theta + y\cos\theta) \end{aligned} \tag{2-3}$$

where $\theta = (n\pi)/K$ and K is the total number of orientations. The scale factor a^{-m} is meant to ensure that the energy is independent of m. This set of functions form a non-orthogonal basis of functions for the multiresolution decomposition.

The non-orthogonality of the Gabor wavelets implies that there is redundant information in the filtered images, and the following strategy is used to reduce this redundancy. Let U_l and U_h denote the lower and upper center frequencies of interest. Let K be the number of orientations and S be the number of scales in the multi-resolution decomposition. Then the design strategy is to ensure that the half-peak magnitude supports of the filter responses in the frequency spectrum touch each other. This results in the following formulas for computing the filter parameters σ_u and σ_v (and thus σ_x and σ_y),

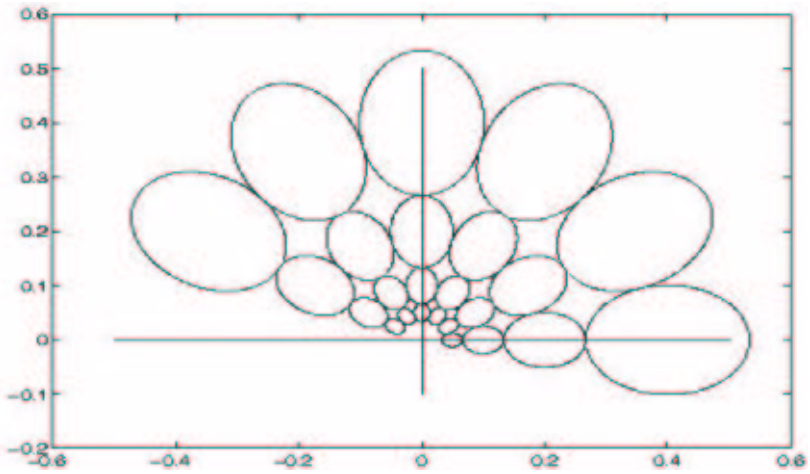


FIGURE 2-2. The contours indicate the half-peak magnitude of the filter responses in the Gabor filter dictionary. The filter parameters used are $U_h = 0.04$, $U_l = 0.5$, $S = 4$ and $K = 6$ [52].

$$a = (U_h / U_l)^{\frac{1}{S-1}} \qquad \sigma_u = \frac{(a-1)U_h}{(a+1)\sqrt{2\ln 2}}$$
$$\sigma_v = \tan\left(\frac{\pi}{2k}\right)\left[U_h - 2\ln 2\left(\frac{\sigma_u^2}{U_h}\right)\right]\left[2\ln 2 - \frac{(2\ln 2)^2\sigma_u^2}{U_h^2}\right]^{-\frac{1}{2}} \tag{2-4}$$

where $W = U_h$ and $m = 1, \ldots, S$. Filtering an image $I(x, y)$ with $g_{mn}(x, y)$ results in a filtered image at a specific direction *n* and a scale *m*,

$$W_{mn}(x, y) = \int I(x, y) g^*_{mn}(x - x_1, y - y_1) dx_1 dy_1 \tag{2-5}$$

where * indicates the complex conjugate.

2.2.2 Extraction of HTD

Given that K orientations and S scales are used for multiresolution decomposition, there are total $S \times K$ filtered images resulted, denoted as $W_{mn}(x, y)$, $m = 1, \ldots, S$ and $n = 1, \ldots, K$. The mean μ_{mn} and the square root of the standard deviation σ_{mn} of the magnitude of the transform coefficients are used to for the component of HTD. Formally, the HTD is a vector formed as following

$$HTD = [\mu_{11} \ \sigma_{11} \ \ldots \mu_{SK} \ \sigma_{SK}] \tag{2-6}$$

Note the double index on the vector elements.

2.2.3 Similarity Measures

To use the HTD for texture image retrieval, it is generally required to define a certain distance measure so that the similarity of textures can be captured by the distances computed from the HTDs. Some of the distance measures include the Euclidean distance, the Mahalalobis distance, the quadratic distance with cross-element correlation and L1 distance. For this descriptor, some previous work [52] suggested a normalized L1 distance. Specifically, consider two image patterns *i* and *j*, the distance between the two patterns is defined to be

$$d(i,j) = \sum_m \sum_n d_{mn}(i,j) \tag{2-7}$$

where

$$d_{mn}(i,j) = \left| \frac{\mu_{mn}^{(i)} - \mu_{mn}^{(i)}}{\alpha(\mu_{mn})} \right| + \left| \frac{\sigma_{mn}^{(i)} - \sigma_{mn}^{(i)}}{\alpha(\sigma_{mn})} \right| \tag{2-8}$$

$\alpha(\mu_{mn})$ and $\alpha(\sigma_{mn})$ are the standard deviation of the respective elements over the entire database and used to normalize the individual vector elements. The detailed evaluation of using this similarity measure in image retrieval can be found in [43], [51], [52].

In recent MPEG-7 evaluation work [53], the following similarity measure is proposed for rotation invariant similarity retrieval,

$$d(i,j) = min_{1 \le t \le K} \left\{ \sum_{m,n} \left[\left| \frac{\mu_{mn}^{(i)}}{\alpha(\mu_{mn})} - \frac{\mu_{mn \oplus t}^{(j)}}{\alpha(\mu_{mn \oplus t})} \right| + \left| \frac{\sigma_{mn}^{(i)}}{\alpha(\sigma_{mn})} - \frac{\sigma_{mn \oplus t}^{(j)}}{\alpha(\sigma_{mn \oplus t})} \right| \right] \right\} \tag{2-9}$$

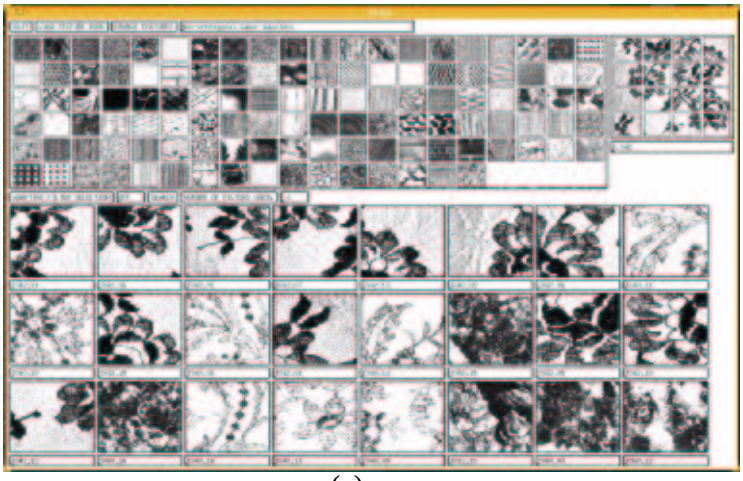
where K is the number of orientations and $\oplus$ denotes a modulo shift in the orientation index. Here we shift the feature vector which is equivalent to a rotation in the spatial domain and then compare the distances between the two patterns under consideration. The minimum of these distances is then used as the distance between pattern i and j for rotation invariant retrieval.

2.2.4 Comparison with other descriptors

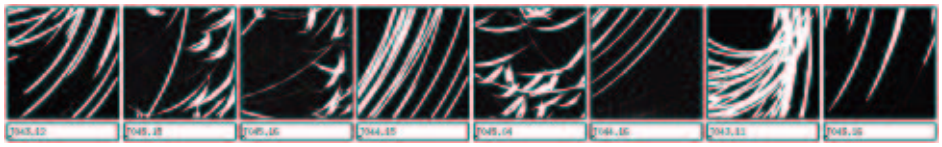
The comparisons are made with the conventional pyramid-structured wavelet transform (PWT) descriptor, tree-structured wavelet transform (TWT) descriptor, and the multiresolution simultaneous autoregressive model (MR-SAR) descriptor.

To compare the effectiveness of using the descriptors in measuring the similarity of texture patterns, we consider a image retrieval application. The Brodatz texture database [14] is adopted as the texture database to provide a quantitative performance evaluation. The Brodatz texture database used in the experiments consists of 116 different texture images. Each of the 512×512 images is divided into 16 non-overlapping subimages, thus creating a database of 1856 texture images. A query pattern in the following is any one of the 1856 patterns in the database. This pattern is then processed to compute the descriptors. The distance $d(i, j)$, where i is the query pattern and j is a pattern from the database, is computed. The distances are then sorted in increasing order and the closest set of patterns are then retrieved. In the ideal case all the top 15 retrievals are from the same large image. The performance is measured in terms of the average retrieval rate which is defined as the average percentage number of patterns belonging to the same image as the query pattern in the top 15 matches. Some retrieval examples are shown in Figure 2-3.

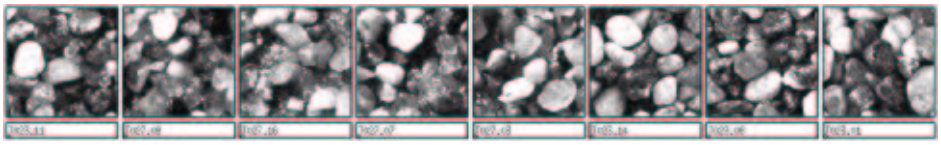
For the HTD descriptor, we set $S = 4$ and $K = 6$, which specifies a 48 dimensional vector. The filter coefficients used for computing PWT are the 16-tap Daubechies orthogonal wavelets (same as the ones used in [17] for the TWT). The 128×128 image pattern is decomposed into three levels ($4 \times 3 = 12$ bands) of the wavelet transform. The mean and standard deviation of the energy distribution corresponding to each of the subbands at each decomposition level are used to construct a 12×2 dimensional descriptor. In [17], decomposition of image subbands at each level is based on energy considerations and this results in a tree structured decomposition where different patterns have different structures. For pattern retrieval applications, it is convenient to have a fixed structure. A fixed decomposition tree can be obtained by sequentially decomposing the LL, LH, and HL subbands. The HH band is not decomposed as this often does not lead to stable features. A three level decomposition results in 52 ($4(1 + 3 + 9)$) subbands. As in the PWT, the mean and standard deviation in each subband are used to construct a 52×2 component descriptor. For the MR-SAR model



(a)



(b)



(c)



(d)

FIGURE 2-3. Texture image retrieval using HTD. Examples shown are some of the difficult patterns to analyze. The average retrieval rate is shown in the parenthesis; (a) Texture retrieval interface with all 116 textures and retrievals for D42, lace (50%), (b) D43, swinging light bulb (54%), (c) D23, beach pebbles (54%), and (d) D91, clouds (25%). The first image in each of (b)-(d) represents the query image. Note that some of the incorrect matches actually look quite similar to the query pattern.

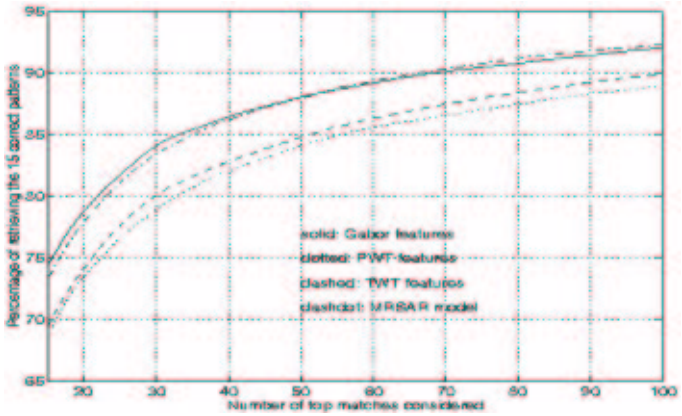


FIGURE 2-4. Retrieval performance according to the number of top matches considered

based descriptor [54], previous work indicate that the MR-SAR features at levels 2, 3, and 4, provide the best overall performance. At each level 5 parameters are computed to represent the texture, thus requiring a total of 15 feature components. The Mahalanobis distance is used to compare the feature vectors.

Figure 7 shows a graph illustrating this retrieval performance as a function of number of top matches considered. Note that only MR-SAR based descriptor performs closely with HTD. But typically, the computation of MR-SAR descriptor is much expensive than that of the HTD [52]. In general, it can be concluded that, as quoted from [43], "... the Gabor feature set proposed in the Alexandria project [52] outperforms other feature sets by a wide margin in the core image benchmark set." .

2.3 Color Descriptors

Color is perhaps the most dominant of the visual features. Using color as a primitive to index an image's content appears in almost all well known image/video information retrieval engines [24], [32]. Since human eye can distinguish thousands of color shades and intensities,

but only two-dozen shades of gray, it is generally recognized that color can be more effective in describing an image's content.

To distinguish colors, different color models are adopted. A color model is a specification of a 3-D coordinate system and a subspace within that system where each color is represented by a single point. The international community CIE provides the standard specifications of the color models. To name a few, some well known color models include RGB, CIE XYZ, CIE LUV, CIE YIQ, etc.. They are utilized by many different applications. For example, RGB color model is widely used by color monitor and color cameras; YIQ model is used for TV broadcast.

Given a color model, each pixel of a color image corresponds to a single point in a 3-D coordinate system. The color information of this pixel is thus characterized by the coordinates of the point. In image/video retrieval applications, a common task is to compare the similarity of images, which requires the characterization of the color information of the whole image. For this purpose, different means have been proposed, including color histogram [76], color coherence vectors [60] and color correlogram [33].

Statistically, color histogram provides the joint probability of the coordinates of image pixels in the 3-D color space. It is invariant to image rotation, translation and viewing axis. Its effectiveness in characterizing global color content has been widely recognized and demonstrated in many image/video retrieval systems [24], [32], [69]. One shortcoming of a color histogram description is that it lacks of information about how the color is spatially distributed. For example, it can not distinguish a single large blue region from a large number of scattered blue pixels. For this reason, descriptors based on the color coherence vectors and the color correlogram are also used for capturing the spatial layout of images. Using color coherence vectors is an approach to incorporate spatial information into the color histogram representation. To compute the color coherence vectors, each pixel is classified as whether or not it belongs to a large uniformly-colored region. The color coherence vectors partitions each histogram bin into two: one representing coherent pixels and the other representing incoherent pixels. The color correlogram characterizes how the spatial correlation of pairs of colors changes with distance in an image.

Despite its shortcomings, the color histogram offers a mature and well tested descriptor for representing color for similarity retrieval.

2.3.1 Color Histogram Descriptor (CHD)

Given an image or a region of an image $I(x, y)$, its color histogram descriptor is an N-dimensional vector

$$\{H(I(x, y), l);\ \ l = 1, \ldots, N\} \tag{2-10}$$

where N is the number of colors and $H(I(x, y), l)$ is the number of pixels in $I(x, y)$ having color l. In constructing the histogram, the 3-D color space is mapped onto a limited number of bins by quantization. For this purpose, the color space is typically quantized. Different color space may be quantized differently. Generally speaking, three approaches can be used to quantize the color space, uniform quantization, non uniform quantization and look-up tables approach. The choice depends on the properties of the chosen color space and the intended applications. For most non uniform color spaces, such as RGB, XYZ, the non-uniform quantization and look-up table approaches are adopted. In this dissertation, we consider the color histogram descriptor constructed from the uniform quantization of the perceptually uniform LUV color space [76].

LUV is a perceptually uniform color space. A color space is perceptually uniform if a perturbation to a component value of the 3-D color vector is approximately equally perceptible across the range of that value. Such perceptual uniformity allows the uniform quantization of the color space. To construct the histogram, each dimension is divided into a certain number of segments with equal intervals. For example, given that each dimension is quantized into 4 segments, the 3-D color space is mapped to a histogram of 64 bins. We denote the color histogram descriptor constructed from such uniform quantization of LUV color space as CHD (Color Histogram Descriptor),

$$CHD = [h(i, 1), h(i, 2), \ldots, h(i, N)] \tag{2-11}$$

where N is the number of bins and $h(i, l)$, $l = 1, \ldots, N$, is the number of pixels in image object i having color l.

To measure the color similarity of two image object *i* and *j*, the L1-norm is typically used to compute the distance

$$d(i,j) = \sum_{l=1}^{N} |h(i,l) - h(j,l)| \quad (2\text{-}12)$$

Figure 2-5 presents two examples of using color histogram for *query by example* application. The color image database is a collection of 969 online images under the “clothing” category. The number of bins of the *CHD* is 64. In Figure 2-5, for each query image (Figure 2-5(a) and (e)), the 3 nearest neighbors are presented. As observed, the retrievals share the similar global color distribution with the query image, which demonstrates the effectiveness of using the color histogram to characterize the color feature. It should be noted that the match of the color distribution relates to the resolution of the quantization. The more bins used to construct the histogram, the better match with the query image can be expected from the retrieval. But as more bins are used to construct the histogram, the cost of indexing the feature space is also higher, due to the increase of the dimensionality.

2.4 Summary

In this chapter, descriptors that characterize two dominant features used in image/video information retrieval, texture and color, are presented. For each feature, after providing an overview of the work in developing the associated descriptors, we present a state of the art descriptor that has been widely used in image/video information retrieval applications. For texture feature, a Gabor filtering based descriptor is introduced. A detailed comparison with other descriptors is provided to demonstrate its effectiveness. For color feature, we introduce a standard color histogram descriptor. Examples of using such descriptor in a color image database are provided .



(a)



(b)



(c)



(d)



(e)



(f)



(g)



(h)

FIGURE 2-5. Example of using CHD for color image retrieval. (a) and (e) are two query images. (b), (c), (d) are the top 3 nearest neighbors with query (a); (f), (g), (h) are the top 3 nearest neighbors with query (e).